

# Chapter 6

## Dynamics of Fractional-Order Chaotic Systems



### 6.1 Lyapunov Exponent Spectrum Calculation Algorithm

The Lyapunov exponent is the average change rate of the exponential separation or convergence of two adjacent orbits in phase space over time, and is used to characterize the extreme sensitivity of the system to initial values as it evolves over time. A positive Lyapunov exponent indicates that the phase volume of the system is expanding and flooding, resulting in the original adjacent orbits in the attractor becoming more and more irrelevant, and the initial value sensitivity of the system is high.

**Definition 6.1** [1] The Lyapunov exponent is the average change rate of the exponential separation or coalescence of two adjacent orbits in phase space over time, and is calculated as

$$Ly = \frac{1}{t_m - t_0} \sum_{k=1}^M \ln \frac{D(t_k)}{D(t_{k-1})} \quad (6.1)$$

where  $M$  is the total iteration number, and  $D(t_k)$  denotes the distance between the two closest points at moment  $t_k$ . The Lyapunov exponent is a quantitative indicator used to distinguish the complexity of sequences in chaotic attractors, and a positive Lyapunov exponent is a necessary condition for the existence of chaos in a nonlinear system.

When the dimension of the system is greater than one, the system is generally described by the set of Lyapunov exponents, i.e., the Lyapunov exponent spectrum. For a multidimensional nonlinear dissipative system, since the system is stable overall, the sum of the Lyapunov exponent spectrum of the system should be less than zero, indicating that the motion of the system in phase space is contracting and stable overall. There are positive and negative values in the Lyapunov exponent spectrum. The negative ones indicate that the phase volume of the system in this direction is contracting and the evolution in this direction is stable, while the positive

ones indicate that the motion of the system in this direction is unstable and the phase volume is expanding and flooding. It results in that the original adjacent orbits in the attractor become uncorrelated after evolution, so that it is difficult to predict the state of the small deviation of the initial state after a long evolution, which reveals the initial value sensitivity of chaos. So the system is chaotic if there is a positive value in the Lyapunov exponent spectrum of the multidimensional system. In applications, the maximum Lyapunov exponent  $Ly_1$  becomes an important indicator to determine whether a system is chaotic or not.

For a continuous chaotic system, let the Lyapunov exponent spectra of the system be  $Ly_1, Ly_2, \dots, Ly_n$  (in descending order), and the necessary conditions for the system to have chaotic attractors are as follows.

- (1) There is at least one positive Lyapunov exponent  $Ly_i > 0$ .
- (2) There is at least one Lyapunov exponent  $Ly_i = 0$ .
- (3)  $Ly_1 + Ly_2 + \dots + Ly_n < 0$ .

According to Chap. 4, the Adomian decomposition algorithm is used to obtain the iterative equation for a fractional-order chaotic system as [2]

$$\mathbf{x}(m+1) = \mathbf{F}_{Ado}(\mathbf{x}(m)) = \begin{cases} F_{1Ado}(\mathbf{x}(m)) \\ F_{2Ado}(\mathbf{x}(m)) \\ \vdots \\ F_{nAdo}(\mathbf{x}(m)) \end{cases} \quad (6.2)$$

The result of each step of the equation is related to the result of the previous step only, so its Jacobian matrix is

$$\mathbf{J}_{sys} = \begin{bmatrix} \frac{\partial F_{1Ado}}{\partial x_1} & \frac{\partial F_{1Ado}}{\partial x_2} & \dots & \frac{\partial F_{1Ado}}{\partial x_n} \\ \frac{\partial F_{2Ado}}{\partial x_1} & \frac{\partial F_{2Ado}}{\partial x_2} & \dots & \frac{\partial F_{2Ado}}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial F_{nAdo}}{\partial x_1} & \frac{\partial F_{nAdo}}{\partial x_2} & \dots & \frac{\partial F_{nAdo}}{\partial x_n} \end{bmatrix} \quad (6.3)$$

Generally, the above Jacobian matrix is obtained using computer symbolic arithmetic functions, such as MATLAB's Jacobian ( $\cdot$ ) function and Maple's Jacobian ( $\cdot$ ) function. The more the number of terms using Adomian decomposition algorithm, the more complex the obtained Jacobian matrix in form. Here, to analyze the dynamics of fractional-order chaotic systems, the Lyapunov exponent spectrum algorithm is designed based on the above discrete iterative equations, Jacobian matrices, and QR decomposition method. The process of calculating Lyapunov exponent based on QR algorithm is [3]

$$\begin{aligned} qr[\mathbf{J}_M \mathbf{J}_{M-1} \dots \mathbf{J}_1] &= qr[\mathbf{J}_M \mathbf{J}_{M-1} \dots \mathbf{J}_2 (\mathbf{J}_1 \mathbf{Q}_0)] \\ &= qr[\mathbf{J}_M \mathbf{J}_{M-1} \dots \mathbf{J}_3 (\mathbf{J}_2 \mathbf{Q}_1)] [\mathbf{R}_1] \end{aligned}$$

$$\begin{aligned}
&= qr[\mathbf{J}_M \mathbf{J}_{M-1} \dots \mathbf{J}_i (\mathbf{J}_{i-1} \mathbf{Q}_{i-2})][\mathbf{R}_{i-1} \dots \mathbf{R}_2 \mathbf{R}_1] \\
&= \dots \\
&= \mathbf{Q}_M [\mathbf{R}_M \dots \mathbf{R}_2 \mathbf{R}_1] = \mathbf{Q}_M \mathbf{R}
\end{aligned} \tag{6.4}$$

where  $qr[\cdot]$  represents the QR decomposition function, and  $\mathbf{J}$  is the Jacobian matrix of the discrete iterative system, and the Lyapunov exponent spectrum of the system is

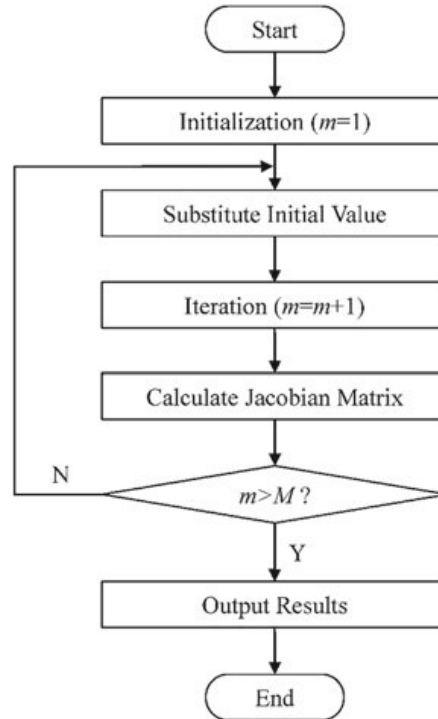
$$Ly_k = \frac{1}{Nh} \sum_{i=1}^N \ln |R_i(k, k)| \tag{6.5}$$

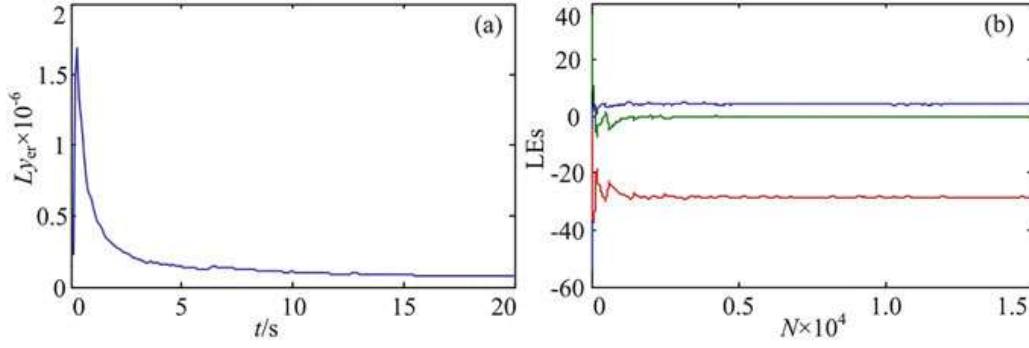
where  $k = 1, 2, \dots, n$  (system dimension).  $N$  is the maximum number of iterations, and  $h$  is the iteration time step. Its calculation flowchart of is shown in Fig. 6.1.

Taking the fractional-order Lorenz chaotic system as an example, we analyzed the convergence, accuracy, and parameter selection of the algorithm. When  $q = 1$ , the sum of Lyapunov exponent spectrum of the system is  $\Delta_{\text{sys}} = -a + d - b = -18$ . Define the global error of the Lyapunov exponent spectrum of the fractional-order Lorenz system as

$$Ly_{er} = \left| \sum_{i=1}^3 Ly_i + 18 \right| \tag{6.6}$$

**Fig. 6.1** Flowchart of Lyapunov exponent spectrum algorithm for fractional-order chaotic systems





**Fig. 6.2** Lyapunov exponent spectrum based on Adomian decomposition algorithm. **a** Global error; **b** the Lyapunov exponent spectrum of the fractional-order Lorenz system

The simulation results are shown in Fig. 6.2a, and it shows that this Lyapunov exponent spectrum calculation algorithm has high accuracy. When  $q = 0.96$ , the convergences of the three Lyapunov exponents are shown in Fig. 6.2b, which shows that the Lyapunov exponent spectra algorithm for fractional-order chaotic systems has good convergence and accuracy.

## 6.2 Dynamical Characterization of a Family of Fractional-Order Lorenz Systems

### 6.2.1 Fractional-Order Lorenz Chaotic System

Fractional-order Lorenz chaotic system ( $D_{t_0}^q x_1 = a(x_2 - x_1), D_{t_0}^q x_2 = cx_1 - x_1x_3 + dx_2, D_{t_0}^q x_3 = x_1x_2 - bx_3, a = 40, b = 3, c = 10$ ) has two parameters,  $d$  and the order  $q$ . Here, we analyze its dynamics as the parameters change.

#### (1) Dynamical properties for $d = 25$ and $q$ varying

Let the step size of the fractional order  $q$  be 0.0005 and the range of variation be in  $[0.75, 1.0]$ . The corresponding Lyapunov exponent spectrum and bifurcation diagram are shown in Fig. 6.3. It can be seen that when  $0.813 \leq q < 1$ , the system is mostly chaotic and there are also some periodic windows. When  $q = 0.813$ , the attractor of the system is shown in Fig. 6.4. When  $q < 0.813$ , the system is divergent. So the minimum order of the system to produce chaos is 2.439. The maximum Lyapunov exponent can characterize the system complexity to some extent, and the larger the maximum Lyapunov exponent, the higher the complexity of the system. From Fig. 6.3a, we found that the maximum Lyapunov exponent of the fractional-order Lorenz system decreases with the order  $q$  increasing, i.e., the system complexity tends to decrease with the order  $q$  increasing. It indicates that the fractional-order Lorenz system has better practical application than the integer-order Lorenz system.

#### (2) Dynamical properties for $q = 0.96$ and $d$ varying